

Hooke's law and simple harmonic motion

Key Points

Simple harmonic motion (SHM) is from the differential equation

$$\ddot{x} = -\omega^2 x$$

for constant ω

Hooke's law gives that the tension in a spring body is

$$T = kx$$

Where $k = \frac{\lambda}{l_0} = \frac{EA}{l_0}$ is the stiffness constant and x is the displacement from equilibrium

Basic Skills

Q1. For the differential equation $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} - 18y = \cos 2x$

a) Solve the homogeneous version

b) Find the particular integral

c) Find the general solution

d) Find the particular solution when $y(0) = 0, \frac{dy}{dx}(0) = 0$

Q2. Write down the relationship between acceleration and displacement

Q3. Convert the expression $4 \cos \theta + 5 \sin \theta$ into $R \sin(\theta + \alpha)$ and hence state the magnitude and the phase shift

Q4. Find the general solution to simple harmonic motion

Competency Skills

Q5. Consider the spring-mass oscillator is in equilibrium when extended e past natural length

a) Using Hooke's law which states: "The tension in a spring body is proportional to the extension past natural length" to show that $ke = mg$

b) Given the mass is then extended an additional x past the point of equilibrium show that the motion of the mass must obey the differential equation $\ddot{x} = g - \frac{k}{m}(e + x)$

c) Hence show why the motion is equivalent to simple harmonic motion showing $\omega = \sqrt{\frac{k}{m}}$

d) Hence find the motion of a spring-mass oscillator with stiffness 50 Nm^{-1} and mass 2 kg which is initially at rest and extended 1.5 m beyond equilibrium

Q6. Find the maximum an object obeying simple harmonic motion is displaced if $\omega = 4$ if it initially travelling at 0.4 ms^{-1} and extended 50 cm beyond equilibrium

Advanced Skills

Q7. Find the angular velocity of a mass on a pendulum for which the angular displacement is given by the differential equation $\ddot{\theta} = -\frac{g}{l}\theta$ where the length of string $l = 2$ and the initial displacement is zero with an initial velocity of 0.4 rad/s

Q8. Explain why simple harmonic motion does not usually occur in real life and suggest how the model can be changed in order to get solutions which more represent real world oscillations

Q9. For $k = \frac{\lambda}{l_0}$ what does the constant λ represent for the spring

Q10. State the amplitude, period and phase shift for sine for an object obeying simple harmonic motion in terms of the constant ω and initial displacement x_0 and initial velocity v_0

Extension

Q11. For the second order ODE, what requirements are necessary so the solution tends to zero at infinity

Hint: Consider the possible solutions to the ODE